Implementation Tools:
Importing Documents with DTW

SAP Business One
Version 10.0

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Welcome to the course on importing documents using the Data Transfer Workbench (DTW). This course should be useful to anyone using DTW for the first time.

Objectives

On completion of this course you will be able to:

- Explain how to import documents and reference other documents as a base
- Describe the relationship between a document's header and row details and how this is represented in DTW using the *Documents* and *Document_Lines* templates
- State the potential effect of importing a document on inventory and the general ledger

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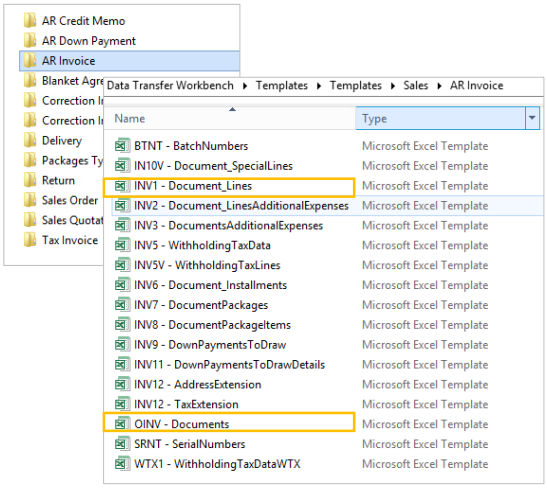
- State the potential effect of importing a document on inventory and the general ledger

Importing Documents

Template Structure for Documents



Most marketing documents (and some inventory and production documents) are based on the same DI API business object:

- *Documents* object for the document header
- *Document_Lines* child object for the document rows

In DTW, the predefined templates follow the same format:

- *Documents* template for the header
- *Document_Lines* template for the rows
- Other child templates for importing text and subtotal rows, additional expenses (freight, insurance, etc.), taxes, and serial number/batch information for the document

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Most marketing documents (and some inventory and production documents) are based on the same DI API business object:

- The *Documents* business object represents the header data of a document
- The *Document_Lines* is a child object of the Documents object and represents the rows of a document

In DTW, the predefined templates follow the same format. There is a *Documents* template for the header data and a *Document_Lines* template for the row data.

In the Templates folder for each document type (A/R Invoice, Goods Return, etc.), you can find the *Documents* and *Document_Lines* templates plus any additional child templates depending on the document type, such as:

- Text and subtotal rows
- Additional Expenses (freight, insurance and other charges). Note that there are two templates for additional expenses. You can import additional expenses at the row level or at the document header level.
- Withholding tax, if applicable
- Batch Numbers
- Serial Numbers

Document Numbering

- You can import a document with its legacy document number, or use SAP Business One automatic document numbering
  - If the *HandWritten* field is blank or tNO, SAP Business One will assign document numbers from the current numbering series
  - To retain legacy document numbers, enter the actual document number and 'tYES' in the *HandWritten* field

OINV - Documents1

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW

	A	B	C	D	E	F	G	H	K	L	N	R
1	DocNum	DocEntry	DocType	HandWritte	Printed	DocDate	DocDueDa	CardCode	NumAtCar	DocTotal	DocCurren	Comments
2	DocNum	DocEntry	DocType	Handwrtten	Printed	DocDate	DocDueDate	CardCode	NumAtCard	DocTotal	DocCur	Comments
3	1		dDocument_Items	tNO	tNO	20180925	20181018	Cust001	Ref001	1500		New
4	2		dDocument_Items	tNO	tNO	20180926	20181019	Cust002	Ref002	23,050		New
5	3		dDocument_Items	tNO	tNO	20180927	20181020	Cust003	Ref003	99.95		New
6	3125		dDocument_Service	tYES	tYES	20180715	20180815	Cust004	Ref004	156.9		Based on order
7	3126		dDocument_Service	tYES	tYES	20180716	20180816	Cust005	Ref005	2000		Based on order
8	3127		dDocument_Service	tYES	tYES	20187917	20180817	Cust006	Ref006	4280		Based on order

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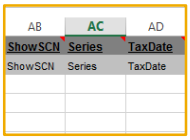
You can import a document with its legacy document number, or use SAP Business One automatic document numbering.
To use SAP Business One numbering, enter any number in the *DocNum* column (column A). The system will assign the next document number in the current numbering series.

To retain the actual document numbers when importing legacy documents, enter the real document number in the *DocNum* column then enter the value 'tYES' in the *HandWritten* field. Documents with manual numbers are considered printed when they are imported.

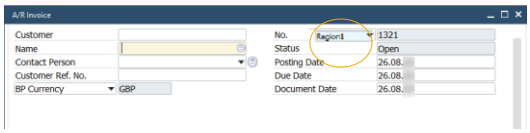
Document Series

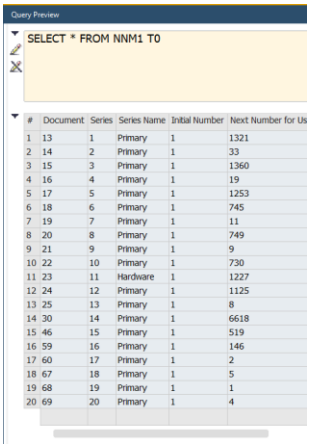
- You can optionally specify a numbering series in the template (in the *Series* column)



AB	AC	AD
ShowSCN	Series	TaxDate
ShowSCN	Series	TaxDate

- The numbering series is an integer assigned by the system whenever you create a new numbering series





#	Document	Series	Series Name	Initial Number	Next Number for Us
1	13	1	Primary	1	1321
2	14	2	Primary	1	33
3	15	3	Primary	1	1360
4	16	4	Primary	1	19
5	17	5	Primary	1	1253
6	18	6	Primary	1	745
7	19	7	Primary	1	11
8	20	8	Primary	1	749
9	21	9	Primary	1	9
10	22	10	Primary	1	730
11	23	11	Hardware	1	1227
12	24	12	Primary	1	1125
13	25	13	Primary	1	8
14	30	14	Primary	1	6618
15	46	15	Primary	1	519
16	59	16	Primary	1	146
17	60	17	Primary	1	2
18	67	18	Primary	1	5
19	68	19	Primary	1	1
20	69	20	Primary	1	4

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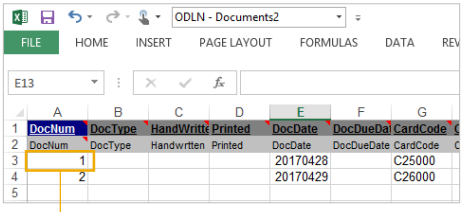
If you want to use a different numbering series for the documents, you can specify the numbering series in the DTW template. Make sure you define the document numbering series before you import the records.

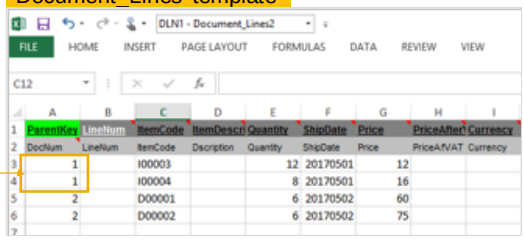
In the *Series* column enter the generated number for the series. This is an integer assigned by the system whenever you create a new numbering series. To see this number, run a query on the NNM1 table and use the Series value that corresponds to the series name. You can also see this value in *System Information* when you hold your mouse over the series number in a document.

Parent and Child Templates

Documents template



Document_Lines template



- Each row in the parent template represents a document
- Each document can have one or more rows in the child template
- To relate the child and parent templates, the parent key field in the child template will link back to the primary key (column A) in the parent template
- For documents, the parent key is the document number.

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Each row in the parent template represents a separate document. Each document in the parent template can have one or more rows in the child template.

The rows in the *Document_Line*s template are linked to their respective row in the parent template by the value in column A (the primary key). For example, if the *DocNum* value in the *Documents* template is 1, each row that belongs to the document will have the ParentKey value 1.

Effects of Importing Marketing Documents

- When you import marketing documents using DTW, postings are made depending on the document type, to inventory and to various accounts in the general ledger
- Postings are identical to manually created documents

	Inventory	BP Balance	A/R or A/P Balance	Inventory Accounts
Sales or Purchase Quotations				
Sales or Purchase Order	●			
Delivery/ Goods Receipt PO	●			●
A/R or A/P Invoice		●	●	
A/R or A/P Invoice (with no preceding documents)	●	●	●	●

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When you import marketing documents using DTW, be aware that postings are made to the general ledger accounts. These postings are identical to postings when the documents are manually created in the application, and will affect inventory values if a continuous stock system is running. As an example:

- An imported sales order will commit (but not remove) stock from inventory. An imported purchase order will commit stock into inventory.
- An imported delivery for the sales order will affect the in stock quantity and will make postings to inventory-related accounts in the general ledger, including the valuation of inventory if continuous stock is in use
- An imported A/R invoice for the delivery will post to the business partner account, as well as revenue and tax-related accounts
- An A/R invoice with no preceding documents will make all the postings and stock changes that would have been made by the preceding sales order/delivery documents.

Open sales and purchase quotations do not affect accounting or inventory levels and can be safely loaded via the DTW.

Opening Balances Considerations

DTW can be an option for importing opening balances from a legacy system:

- Item costs and quantities using Goods Receipts documents
- Business partner balances using A/R and A/P invoices

Care must be taken to ensure that inventory or business partner balances are not “double-counted”

	Inventory	BP balance	A/R or A/P Balance	Inventory accounts
Goods Receipt	●			●
Item type A/R or A/P Invoice (with no preceding documents)	●	●	●	●
Service Type A/R or A/P Invoice (with no preceding documents)		●	●	

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DTW can be an option for importing opening balances from the legacy system to a new SAP Business One system:

- Item quantities and costs using the Goods Receipt document template
- Business Partner account balances using the A/R and A/P invoice document templates, as well as partial payments

Note: If you are loading the transactions as part of the opening balances for a new company, you must make sure that the documents you import are not “double counted” in inventory or business partner balances. If an “item type” invoice is imported, and there are no preceding documents such as deliveries or goods receipt POs, the invoice will assume the role of the delivery or goods receipt and will increase or decrease inventory quantity. If perpetual inventory is in use, there will also be an automatic posting to the inventory account.

You can avoid the “double entry” of inventory balances by importing the A/R or A/P invoices as service type documents – this also avoids a change to inventory levels. See the Data Migration Guide in the AIP materials on PartnerEdge for more details on opening balances.

Example: Service Type Invoices

Templates > Sales > AR Invoice
OINV – Documents
INV1 – Document_Lines

Templates > Purchasing > AP Invoice
OPCH – Documents
PCH1 – Document_Lines

OINV

A	B	C	D	E	F	G	H	J
DocNum	DocType	HandWritt	Printed	DocDate	DocDueDate	CardCode	CardName	NumAtCa
DocNum	DocType	Handwrtten	Printed	DocDate	DocDueDate	CardCode	CardName	NumAtCard
506	dDocument_Service	tYES		20160116	20160207	C110040		

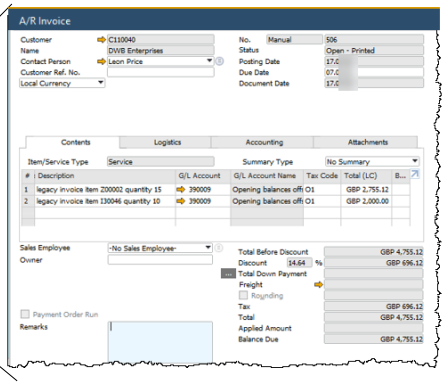
Service Type

Manual document numbering

Keep original dates for aging

INV1

A	B	D	F	H	I	K	R
ParentKey	LineNum	ItemDescription	ShipDate	PriceAfterV	Currency	DiscountP	AccountCode
DocNum	LineNum	Dscription	ShipDate	PriceAfVAT	Currency	DiscPrcnt	AcctCode
506	0	legacy invoice item Z00002 quantity 15		2755.12			390009
506	1	legacy invoice item I30046 quantity 10		2000			390009



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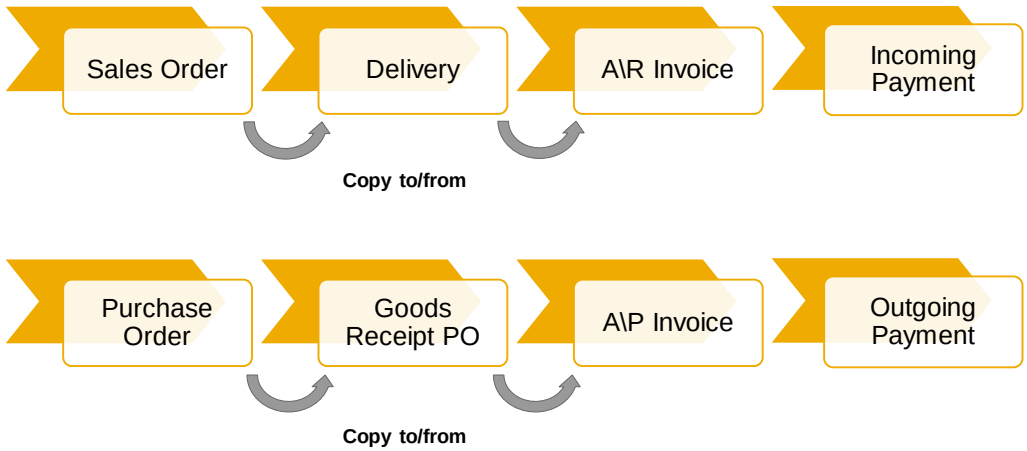
- The DTW templates for A/R and A/P invoices are in the Sales or Purchasing template folders. The templates use the naming convention of the database table. For an A/R invoice the header table is OINV and the row table is INV1. For an A/P invoice the header table is OPCH and the row table is PCH1.
- In the DTW template, indicate a service type document in the DocType column by entering *dDocument_Service*.
- If you want to use the legacy document numbers, enter the number in the *DocNum* column and set the *HandWritten* column value to 'tYes' to indicate manual numbering. In the example the *DocNum* is 506.
- In the row (child) template, you can enter information to describe the original item invoice in the *ItemDescription* column.
- Keep original dates from legacy invoice to maintain correct aging. Note: make sure that you have defined posting periods in SAP Business One so that postings can be made with the original due dates from the legacy system. And make sure these periods are open.
- The imported service invoice will update the business partner balance and the A/R or A/P control accounts. To avoid an update to revenue or expenses accounts, use a temporary clearing account (opening balances offset) as the account code. The balances for the revenue and expenses accounts can be transferred later when you enter the opening balances for G/L accounts.

Importing Related Documents

To maintain the links between related documents for a transaction, you can enter base document information when you import a document.

Importing Related Transactions



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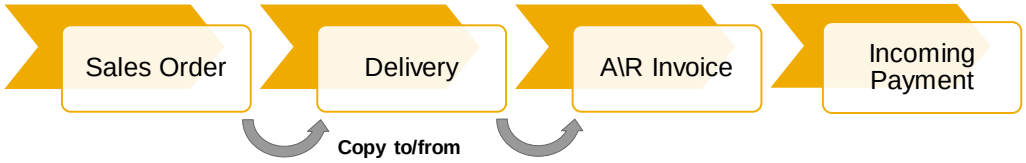
When documents are created manually in SAP Business One, they are linked using the *Copy to/from* feature.

You can replicate the copy to/from in DTW by entering the base document information when you import the related document.

Note: As an alternative to importing each related document using DTW, for sales documents you can use the *Document Generation Wizard.* You import the starting document (e.g., sales order) using DTW and then run the wizard in SAP Business One to generate the related documents. However, if there are significant changes to the item rows or dates, you should import the related transactions using DTW since you cannot make these changes using the *Document Generation Wizard*.

Base and Target Document Links

- When you import a document that is based on another document, you must provide the base document number, type and row in the DTW template
- The new document will close the base document



	Sales Order	Delivery	A/R Invoice
Document Number	281	300	350
Base Document Number	-	281	300
Base Document Type	-	Sales Order	Delivery
Base Row – 1st row Base Row – 2nd row	-	0 1	0 1

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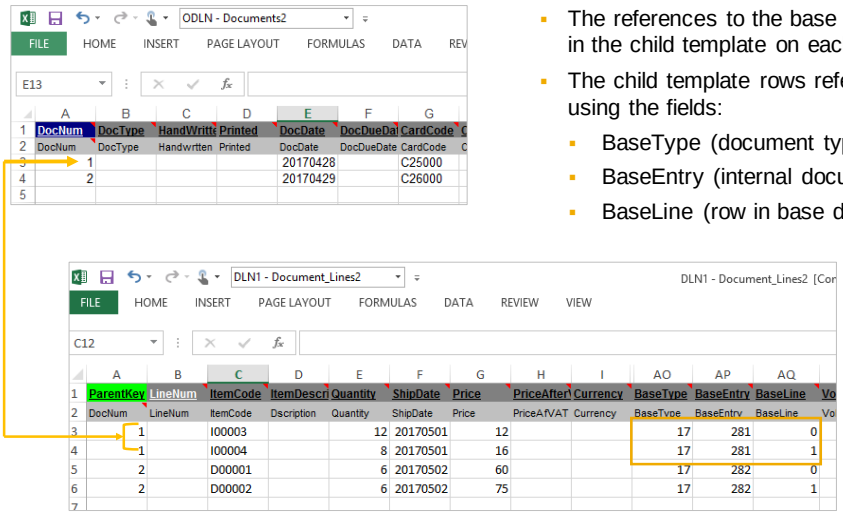
When you import a document that is based on another document, you must provide details about the base document so that the system can link them together. There are three columns in the template used for this:

- Base document number (the document number of the preceding document in the transaction)
- Base document type (the document type: each document type is identified with a unique integer value)
- Base document row (starts at 0 for row 1)

When you import a related transaction using DTW, it closes the base document.

Note: you do not have to enter all row details for a related transaction. Information will be automatically copied from the base document (like the copy/to). The exception to this is additional expenses, such as freight and insurance, which are not copied and must be imported for each document.

Specifying Base Document References in the Template



- The references to the base document are specified in the child template on each row
- The child template rows reference a base document using the fields:
  - BaseType (document type)
  - BaseEntry (internal document number)
  - BaseLine (row in base document)

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The references to the base document are specified in the child template on each row using the BaseType, BaseEntry and BaseLine fields:

- The BaseType is the document type, for example, 17 indicates a sales order.
- The BaseEntry is the internal document number of the base document. This is the auto-assigned internal number (*DocEntry* field) and not the document number (*DocNum* field). The internal number initially starts at 1 for each document type, and is incremented whenever a new document is imported or added to the system. For the most part, the two numbers will be identical unless you have used manual numbering. However, you should always verify the internal numbers of the base documents by running a query on the base document table. The base type, internal ID and row are held in the child table (for example, INV1) and not the header table.
- The BaseLine is the row instance in the base document. This is an integer, with 0 for row 1, 1 for row 2, and so on.

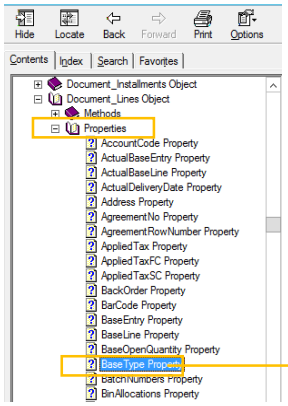
In the example shown, a Delivery document (ODLN) with 2 rows is being importing using DTW. The Delivery is related to a base sales order that has an internal document number 281. The references to the base sales order are entered in the child template for the Delivery:

- The base type is 17 (sales order)
- The base internal document number is 281
- The base line number 0 relates to the first row in the sales order, and base line number 1 relates to the second row.

Note: You do not have to enter all row details for a related transaction. Information will be automatically copied from the base document. The exception to this is additional expenses, such as freight and insurance, which are not copied and must be imported for each document.

Base Document Types

- Base document types are documented in the DI API Objects Reference



BaseType	Document Type
23	Sales Quotation
17	Sales Order
15	Delivery
13	A/R invoice
16	Return
22	Purchase order
20	Goods Receipt PO
18	A/P invoice
14	Sales Credit Memo
19	Purchase Credit Memo
21	Goods Return

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To see the base document types, open the DI API Objects Reference and expand the *Document_Lines* object, then select the *BaseType* property. Click the *BoAPARDocumentTypes* link to see the full list.

Alternately you can run a query in SAP Business One on the document row table to view the object type.

Summary

Key points from this course:

- You can import marketing documents using DTW. You can import a document with its legacy document number, or use SAP Business One automatic document numbering
- Most marketing documents (and some inventory and production documents) are based on the same DI API business object. DTW uses this structure with the document templates:
  - *Documents* object for the document header
  - *Document_Lines* child object for the document rows
- You can import documents that are linked to base documents by specifying the base document information in the child row template
- When you import marketing documents, postings are made to the general ledger accounts. These postings are identical to postings made when the documents are manually created
- DTW can be an option for importing opening balances:
  - Item quantities and costs using the Goods Receipt document template
  - Business Partner account balances using the A/R and A/P invoice document templates
  - Make sure that the documents you import are not “double counted” in G/L account balances.

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Here are some key points to take away from this course:

- You can import marketing documents using DTW. You can import a document with its legacy document number, or use SAP Business One automatic document numbering.
- Most marketing documents (and some inventory and production documents) are based on the same DI API business object. DTW uses this structure with the document templates:
  - *Documents* object for the document header
  - *Document_Lines* child object for the document rows
- You can import documents that are linked to base documents by specifying the base document type, base document internal number, and base row in the child row template
- When you import marketing documents, postings are made to the general ledger accounts. These postings are identical to postings made when the documents are manually created
- DTW can be an option for importing opening balances:
  - Item quantities and costs using the Goods Receipt document template
  - Business Partner account balances using the A/R and A/P invoice document templates

Note: If you are loading the transactions as part of the opening balances for a new company, you must make sure that the documents you import are not “double counted” in inventory or in G/L account balances. You can import invoices as service-type documents to avoid a posting to inventory quantity or inventory value accounts.

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